KAleep-Net: A Kolmogorov-Arnold Flash Attention Network for Sleep Stage Classification Using Single-Channel EEG with Explainability

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Abstract- *Classification of the stages of sleep is crucial in assessing the quality of sleep and diagnosing sleep disorders, including obstructive sleep apnea (OSA), which results in disturbances in sleep architecture. The most reliable method used in the assessment of sleep stages is polysomnography (PSG). Nevertheless, the process of manually scoring sleep is tedious and time-consuming, and also susceptible to in the form of inter-scorer variability. The existing automated approaches based on machine learning as well as deep learning models are mostly based on Euclidean representations of the electroencephalogram (EEG) signals, which are limited in capturing the nonlinear channel interactions, as well as the dynamic changes in the brain signals, especially in the case of patients with OSA. In order to overcome the limitations of the existing approaches, a robust framework for the classification of the stages of sleep based on the electroencephalogram (EEG) signals is proposed, which incorporates the use of nonlinear geometric modeling as well as the use of dynamic feature representations. The proposed framework is based on the use of covariance representations, which are effective in modeling the nonlinear interactions between the channels of the EEG signals. The proposed framework incorporates the use of both static as well as dynamic features in order to effectively represent the brain signals during the stages of sleep. In addition, the proposed framework incorporates the use of a cross-subject adaptation mechanism in order to reduce the intra-class diversity, which makes the proposed framework more robust in nature compared to the existing approaches. The proposed framework is compared with the existing approaches on the publicly available sleep datasets, which shows the competitive performance of the proposed framework while achieving the benefits of robustness as well as computational efficiency compared to the existing approaches. The proposed framework is expected to be useful in real-world as well as clinical environments, as the proposed framework is based on the use of nonlinear geometric modeling, which makes the proposed framework more interpretable compared to the existing approaches.*

I. INTRODUCTION

Sleep is a basic human biological process that is essential in keeping our brain sharp, managing our emotions, balancing our metabolism, supporting our immune systems, and generally promoting our health. While the body is sleeping, several important processes are occurring, including cell repair, memory consolidation, hormonal balancing, and brain organization. Sleep is therefore essential in promoting healthy brain function, preventing diseases such as heart disease, high blood pressure, diabetes, depression, among others, while sleep deprivation can impair learning, job performance, increase the risk of accidents, and increase mortality risk. There is therefore a growing body of evidence that links sleep deprivation with several leading causes of death, thereby establishing sleep insufficiency as a global health threat.

Among the sleep disorders, Obstructive Sleep Apnea (OSA) is notable for its high prevalence and clinical importance, with more than a billion people worldwide suffering from this disorder. OSA is characterized by the obstruction of the upper respiratory airway during sleep, leading to the cessation of breathing or shallow breathing. This disrupts the normal architecture of the sleep pattern and causes excessive daytime somnolence. Sleep stage analysis is important in the diagnosis and treatment of OSA and in the prediction of the outcome. Sleep staging refers to the classification of the entire sleep cycle into different stages based on the recommendations of the American Academy of Sleep Medicine (AASM). The different stages are Wake (W), REM, and the three NREM stages: N1, N2, and N3. Each stage has its own unique neurophysiological characteristics that are reflected by the corresponding electroencephalogram (EEG) pattern. The EEG consists of several types of electrical activity corresponding to the different frequency bands: delta, theta, alpha, sigma, beta, and gamma. For example, the delta band is associated with deep sleep. REM sleep is characterized by the presence of high-frequency activity. In patients with OSA, the abnormal respiratory patterns cause irregular oscillations in the EEG pattern. This makes the sleep stage analysis more complex compared with normal subjects.

Polysomnography (PSG) is still considered the gold standard for monitoring sleep and diagnosing OSA. Here, multiple physiological signals, such as EEG, EOG, EMG, and ECG, are recorded concurrently, while sleep technologists traditionally score 30-second epochs to determine sleep stages. While this method is well-established, it is labor-intensive, time-consuming, expensive, and subject to scorer variability. Furthermore, using multiple signals makes this method less convenient.

To increase the efficiency of sleep stage classification, there is a lot of research activity focused on automated sleep stage classification. Initially, traditional machine learning techniques were used, where features were first extracted using time, frequency, or time-frequency domain representations, followed by classification using Naïve Bayes, SVM, RF, or KNN classifiers. These techniques were found to outperform rule-based methods. However, their performance is highly dependent on feature extraction, which is done by experts. Therefore, using these techniques may not be scalable.

Recently, deep learning techniques have greatly improved automated sleep stage classification. CNNs have shown strong ability to extract hierarchical spectral features, while RNNs and LSTMs have demonstrated their ability to capture temporal dynamics of sleep stages. Attention mechanisms and multi-head self-attention have further improved performance by modeling both within-epoch and between-epoch relationships. However, there are still many challenges to overcome. First, most studies have concentrated on healthy subjects, while patients with OSA have not received much attention. Second, breathing events cause high intra-class variability, while abnormal EEG patterns have been reported. Third, EEG signals have strong non-linear characteristics, while there is high inter-channel covariance. Most traditional ML and deep learning methods have failed to capture these characteristics, where EEG signals have not been considered a vector in flat space. Fourth, deep learning methods require a lot of training data, which is not available for OSA patients. Finally, there is high variability between subjects, where one model trained on one subject may not generalize well to others.

Most existing approaches rely heavily on static feature extraction and do not account for the dynamic behavior of the EEG signal caused by respiratory disturbances. For OSA patients, the changing and arousing events result in complex time trajectories that are beyond the capability of existing feature extraction approaches. Another problem with existing approaches is the unbalanced number of samples for different sleep stages.

Recently, the authors proposed the use of geometric learning theory based on the Riemannian manifold for the analysis of the EEG signal. Instead of using the traditional flat space for feature extraction, the proposed approach uses EEG signal. The proposed approach uses the points defined signal and maps the points onto the Riemannian manifold. This approach is more accurate and robust compared to the existing approaches. The proposed approach also requires fewer training data and is more interpretable compared to the deep learning-based approaches.

Considering the nonlinearity of the EEG signal and the intra-class variation of the OSA patients, and the requirement for robust cross-subject testing, there is a need for developing robust and accurate sleep staging approaches based on the integration of nonlinear geometric modeling and dynamic feature extraction.

The development of robust and accurate automated sleep staging approaches for OSA patients has the potential for significant improvements in the accuracy of the diagnosis and the quality of patient care in the field of sleep medicine.

II. RESEARCH MOTIVATION

1. Sleep is a fundamental and essential process that keeps our bodies healthy, sharp, and emotionally stable. When we are asleep, our bodies repair cells, consolidate memory, control metabolism, and boost our immune systems. Well-rested sleep also keeps our brains working efficiently and our emotions in a balanced state and increases our overall satisfaction with our lives. On the other hand, a lack of sleep or poor sleep quality may lead to serious health issues such as heart disease, diabetes, obesity, depression, and diminished cognitive and attention abilities. In recent times, sleep disorders have become a rising trend due to our hectic and modern lifestyle characterized by excessive stress, irregular working schedules, increased use of electronic media, and a sedentary lifestyle. Therefore, it has become a hot topic in both clinics and laboratories to monitor and study sleep patterns.
2. Amongst various sleep disorders, Obstructive Sleep Apnea Syndrome (OSA) is a major and most prevalent sleep disorder that affects millions worldwide. It is characterized by partial or total obstruction of the upper airways during sleep that leads to pauses in breathing during sleep. These pauses in breathing cause fragmented sleep and reduce oxygen levels in the blood, resulting in repeated awakenings throughout the night. If not treated in a timely manner, it may lead to high blood pressure, heart disease, stroke, and cognitive impairment. Therefore, it is important to diagnose and monitor sleep stages in order to assess the severity of this dangerous sleep disorder.
3. Until recently, polysomnography (PSG), which is a sleep study, was considered the "gold standard" for sleep research. PSG is a procedure for recording and monitoring multiple physiological activities simultaneously, including EEG, EOG, EMG, and ECG. Sleep stages are then manually classified according to the standards of the American Academy of Sleep Medicine. Sleep is divided into five stages: Wake (W), REM, and non-REM sleep stages N1, N2, and N3, with specific EEG characteristics for each.
4. Although PSG is a comprehensive procedure, there are certain limitations associated with PSG. PSG is a costly procedure, as it requires special equipment, skilled technicians, and a sleep laboratory. Manual sleep staging is a labor-intensive procedure, as it requires skilled professionals to visually observe 30-second EEG segments over a period of hours. This is a clear indicator of the importance of automated sleep staging, as it can greatly enhance the speed of sleep staging and reduce the dependency on manual sleep staging.
5. Machine learning and deep learning techniques have been extensively investigated for automated sleep staging from EEG signals. Typically, a wide range of features is extracted from the EEG signal, and then a specific classifier is applied. Common classifiers include SVM, Random Forest, and KNN. These approaches heavily rely on domain knowledge for extracting relevant features from the EEG signal. These features may be either time-domain, frequency-domain, or time-frequency domain features. These approaches are known for their limitations, as it is extremely difficult for a classifier to generalize across different datasets and people, as EEG signal patterns may vary from person to person and may also contain noise from the environment.
6. Deep learning is a promising alternative approach for sleep staging, as it can learn hierarchical representations from raw EEG signals. Convolutional neural networks (CNNs) can learn spatial and spectral features from the EEG signal, while recurrent neural networks (RNNs) and their variants, such as LSTM, can learn temporal dependencies among sleep stages. Recently, attention-based models were also introduced for sleep staging, where attention is paid to specific periods of the EEG signal.

Another challenge is the spectral diversity of EEG signals. Each sleep phase has a distinct frequency signature, with delta waves dominating deep sleep, theta waves dominating lighter sleep, and a mix of frequencies during REM sleep. The traditional approach may fail to simultaneously grasp both the finer details and the broader spectral trends, and the transitions from one phase to another may go unnoticed. Furthermore, EEG signals are non-stationary, noisy, and individual-dependent, which only adds more complexity to the task of feature extraction and accurate classification.

The major objective of the current study is, therefore, to develop an effective and efficient deep learning framework for accurate sleep stages classification from single-channel EEG, while also bypassing the limitations of existing approaches. This is achieved by integrating multi-spectral feature extraction with advanced temporal modeling and attention mechanisms, which can result in improved classification accuracy, reduced computational cost, and a more effective solution for sleep monitoring.

The challenges, therefore, necessitate the development of more advanced frameworks that can learn multi-scale features from EEG, model temporal relationships, and also be computationally efficient. Thus, considering the limitations of traditional approaches, integrating multi-spectral feature extraction with advanced deep learning models may provide a more effective solution for accurate sleep stages classification. Multi-scale features can learn both the fast and slow changes in the signal, which are essentially the high-frequency and low-frequency oscillations, respectively, and are critical for distinguishing one sleep phase from another.

PROPOSED SYSTEM

This study proposes a framework for sleep stage classification using a single-channel EEG signal, aiming to improve the accuracy, efficiency, and interpretability of sleep monitoring. This framework, called KAleep-Net (Kolmogorov–Arnold based Sleep Network), integrates multi-spectral feature extraction, nonlinear representation learning, and temporal attention, facilitating. This framework is designed to address common issues associated with conventional sleep stage classification methods, such as limited spectral range, inadequate modeling of temporal dependencies, and high computational complexity.

This framework processes 30-second EEG signals, which is consistent with AASM guidelines. Each signal is a 1D signal of 3000 data points, sampled at 100 Hz. This framework is composed of a deep learning architecture, which is divided into two stages: feature extraction, temporal modeling, and classification. The framework is composed of two main parts:

1) Multi-Spectral Feature Pipeline (MSFP)
2) Temporal Attention Network (TAN)

This framework is designed to enable dual modeling of spectral and temporal dependencies of EEG signals, which is essential for accurate sleep stage classification.

A. Multi-Spectral Feature Pipeline

Sleep EEG signals have complex frequency characteristics, which vary depending on the sleep stages. For instance, deep sleep is characterized by low-frequency delta waves, while lighter stages and REM sleep have higher frequency components. To fully understand these characteristics, fine-scale and coarse-scale spectral features need to be considered. To address this, the KAleep-Net uses a dual-branch MSFP.

The first branch is called the "Fine-Scale Feature (FSF) Block," which is designed to extract high-resolution spectral features. To achieve this, convolutional neural network (CNN) blocks are utilized, which have convolutional, pooling, and Kolmogorov–Arnold Network (KAN) transformation blocks. Unlike conventional CNN blocks, each CNN layer is followed by a KAN transformation, which is essential for accurate nonlinear representation learning. Nonlinear representation learning is essential to fully capture the complex dependencies of EEG signals, which is not possible using conventional CNN blocks.

The second component is called the Coarse-Scale Feature (CSF) block and tries to capture general time-based patterns and low-frequency oscillations in the EEG signals. It uses convolutional layers with larger kernel sizes to look at longer chunks of the input signals. Like the FSF block, this CSF block also uses KAN layers for nonlinear feature learning. With both fine and coarse spectral features learned by this model, it achieves a full-minded representation of EEG characteristics.

Finally, after learning the features from both branches, a unified feature map is obtained by combining the outputs from both branches. This unified representation contains both high and low frequencies and will be helpful in recognizing patterns from different stages of sleep.

B. Temporal Attention Network

Sleep stages are sequentially dependent because they change with time. It is important to capture temporal dependencies between different segments of the input EEG signals for accurate sleep stage labeling. To meet this requirement, a Temporal Attention Network (TAN) is designed to examine temporal relationships in the input sequence.

The TAN uses a Temporal Sequencing Network as its first component. It uses a Bi-LSTM model in this component because it is a powerful tool for understanding sequential data and temporal dependencies between different segments in a sequence. Bi-LSTM stands for Bidirectional Long Short-Term Memory and allows the model to look at both past and future contexts in a sequence. This will be helpful in understanding how sleep stages change gradually from one stage to another.

Finally, a Flash Attention mechanism is used in this model to reinforce learning in the temporal feature representation. Attention mechanisms are helpful in this model because they allow the model to weight different segments in a sequence differently and look at more informative segments in the input sequence. Flash Attention is especially helpful in this model because it increases efficiency in model training and still allows for accurate attention modeling.the output of the Flash Attention module is a condensed representation that summarizes the most relevant temporal features within the EEG sequence. This representation is then forwarded to the classification layer.

C. Sleep Stage Classification

Lastly, the sleep stage classification module receives the EEG signal in epoch form and classifies it as any of the five stages. The output of the Temporal Attention Network is fed into a dense layer followed by a softmax layer, resulting in probabilities for Wake (W), N1, N2, N3, and REM stages. The learning process occurs during the training phase by minimizing the categorical cross-entropy loss function. This minimizes the difference in probabilities for the predicted stages and the actual stages.

D. Advantages of the Proposed System

The KAleep-Net system has several advantages over traditional sleep stages classification systems. The multi-spectral feature pipeline ensures that the system is able to capture a wider spectrum of features in the EEG signal. This improves the system's ability to differentiate between the stages. The Kolmogorov-Arnold Network improves the system's ability to learn complex features in the EEG signal. The temporal attention module ensures that the system is able to efficiently handle sequential relationships in the signal without increasing computational costs.

In summary, this system is a compact solution for sleep stages classification using single-channel EEG.

III. OBJECTIVES

The main goal behind this work is to design a fast and reliable automated system that can efficiently classify sleep stages based on input from EEG signals. Sleep stage classification is a significant process in diagnosing sleep disorders and assessing the quality of sleep in a patient. However, it is a labor-intensive process that requires specialized expertise and is time-consuming in its current form. Therefore, we propose a framework that uses deep learning techniques to efficiently classify sleep stages from EEG signals.

The main goals behind this work are as follows:

1) Construct an automated framework for classifying sleep stages.

The main goal behind this is to construct a reliable deep learning model that can efficiently classify sleep stages from input EEG signals and minimize the need for manual scoring by experts.

2) Extract significant features from a single-channel EEG.

The main goal behind this is to design a system that extracts significant features from input EEG signals by considering both high and low frequencies. This will help in maintaining significant information that corresponds to different sleep stages.

3) Utilize advanced architectures based on neural networks for better feature representation.

The study also aims to combine convolutional neural networks with Kolmogorov-Arnold Network (KAN) transformation to enhance nonlinear feature learning, which can help the model detect intricate patterns in the EEG signals that might otherwise remain elusive to other models.

4) Model Temporal Dependencies in EEG Signals

Sleep has a particular sequence of stages that occur in a temporal manner as we move through a sleep cycle. Hence, there needs to be a part of this system that can model this temporal relationship between sleep stages, which can be a Bidirectional LSTM model along with an attention mechanism.

5) Improve Computational Efficiency

This system aims to retain high accuracy in classifying sleep stages while reducing training times as well as overall computation requirements, which can help in real-time sleep monitoring scenarios.

6) Evaluate Performance on Benchmark Sleep Datasets

Finally, this system aims to classify sleep stages reliably using publicly available sleep datasets such as Sleep-EDF and SHHS, along with appropriate metrics such as accuracy, F1 score, and Cohen's Kappa score.

Therefore, it can be seen that this system aims to develop a sleep stage classifier that can help clinicians diagnose sleep disorders more effectively.

IV. ALGORITHM

The KAleep-Net framework for sleep stages classification from a single-channel EEG signal starts with raw EEG signal data, which is recorded from polysomnography (PSG) measurements. EEG signal measurements are often noisy, as there may be eye movement, muscle, and environmental noise. To reduce noise from the raw EEG signal, preprocessing is carried out. A band-pass filter is applied, and the frequency range is set from 0.5 Hz to 45 Hz. This range includes delta, theta, alpha, beta, and gamma frequency bands, which correspond to different stages of sleep. The raw EEG signal is divided into 30-second intervals, as recommended by the American Academy of Sleep Medicine (AASM) for sleep stages. Each of these intervals includes a constant number of samples, for example, 3000 samples with a sampling frequency of 100 Hz.

Next, the epochs are subjected to normalization to standardize the amplitude distributions across subjects and sessions, thus increasing the stability of the training process.

After preprocessing and segmentation, the epochs are fed into the KAleep-Net's Multi-Spectral Feature Pipeline (MSFP). The sleep-related EEG patterns comprise a variety of frequency bands. Delta waves are present in deep sleep stages or N3 sleep, theta waves in lighter stages, and a mix of frequencies in REM sleep. Thus, to incorporate this variety of spectral features, two feature extraction pathways are used in the KAleep-Net's Multi-Spectral Feature Pipeline.

The first pathway is the Fine-Scale Feature (FSF) block. This pathway is designed to incorporate high-frequency features using 1D convolutional layers with small kernel sizes. This is useful for identifying high-frequency features such as sleep spindles, K-complexes, and other transient EEG features that signal transitions in sleep stages. A Kolmogorov-Arnold Network (KAN) transformation is applied after each convolutional layer in the FSF block. This is useful for representing features nonlinearly. This is important for representing features nonlinearly as sleep staging is a nonlinear process.

The second component is called the Coarse-Scale Feature (CSF) block and explores more general temporal patterns in the EEG signal. It uses larger convolutional filters to look at longer time intervals and detect low-frequency oscillations such as delta waves. These slow waves are particularly important in recognizing deep sleep stages. Like the FSF block, CSF uses KAN layers to further increase nonlinear feature modelling capacity.

Combining both FSF and CSF blocks gives us two views of the input EEG signal from different perspectives. One looks at fine-grained spectral information, and the other looks at more general temporal information. The outputs from both blocks are concatenated in a unified form called a multi-spectral feature representation.

Now that we have our features, we use a Temporal Attention Network (TAN) to process our combined sequence. Sleep stages have a sequential pattern and change throughout a night's sleep. Therefore, it is important to capture temporal relationships in order to understand transitions between stages. In order to satisfy this requirement, TAN contains a Bidirectional Long Short-Term Memory (BiLSTM) module.

The BiLSTM model processes the sequence of features in both directions. This gives us two perspectives on the same problem. The importance of the two perspectives lies in identifying patterns and stages of time. This contributes to better accuracy of the classification of the algorithm. Next, the weighted attention on the feature representation moves on to the last stage of classification. In this stage, the features are projected through a fully connected layer.

- Collect the raw EEG signal from the PSG sleep recording data.
- Apply any necessary preprocessing steps to the collected data.
- Segment the collected EEG signal into 30-second epochs.
- Normalize the collected EEG signal.
- The collected EEG epoch is passed through the Fine-Scale Feature (FSF) block. The FSF block uses CNN and KAN to extract the features.
- The collected EEG epoch is passed through the Coarse-Scale Feature (CSF) block. The CSF block uses CNN and KAN to extract the features.
- Concatenate the outputs of the FSF and CSF blocks to generate a unified multi-spectral feature representation.
- Next, we use a Bidirectional LSTM to understand the temporal relationship between features.
- Subsequently, we use the Flash Attention mechanism to identify the important temporal features in the sequence.
- We then use a fully connected layer with soft max activation to obtain the predicted sleep stage label for each EEG segment.
- This entire process is repeated to obtain a sequence of sleep stages.

V. FLOW DIAGRAM

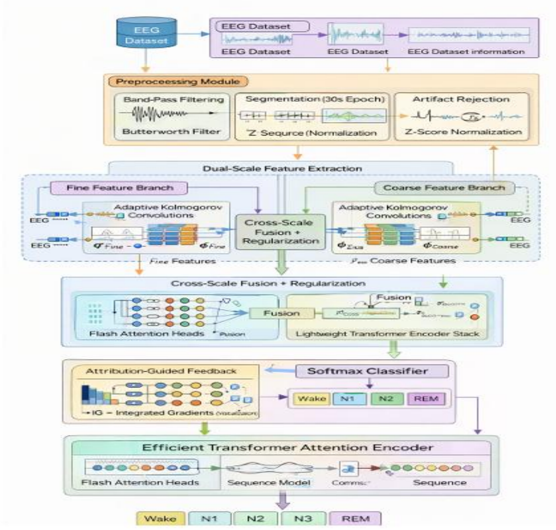
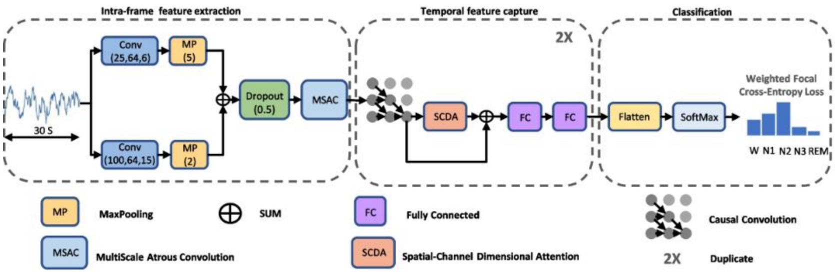


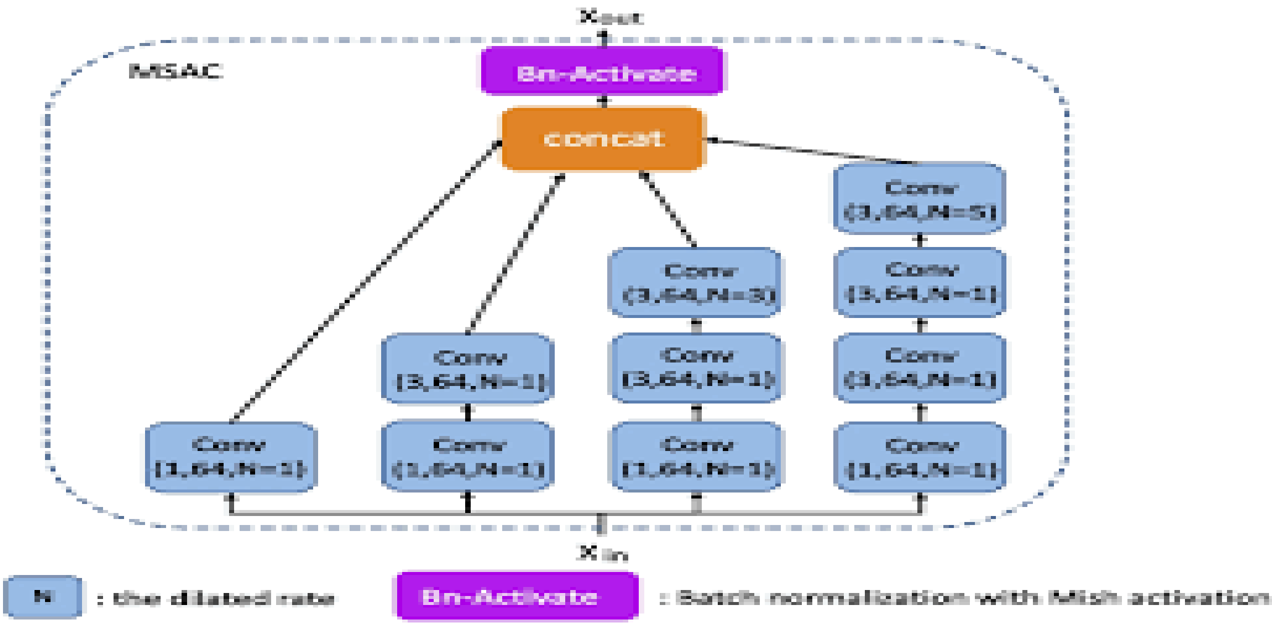
Fig 6.1 System Flow of the Sleep Architecture Diagram

The above diagram illustrates the entire workflow for the proposed sleep stages classification system using the EEG signal classification method. The system begins with an EEG dataset, which is raw brain signal recordings collected during sleep studies along with relevant dataset information. This signal is a representation of the brain's electrical activity and is used for analyzing the various stages of sleep. Before any classification is performed, the raw EEG signal is subjected to preprocessing to enhance the quality of the signal for effective feature extraction. Here, in the preprocessing phase, the raw EEG signal is subjected to a band-pass Butterworth filter to eliminate noise and confine the frequency bands to those relevant for sleep stages. The signal is then segmented into epochs of 30 seconds each, as recommended in the American Academy of Sleep Medicine (AASM) standard for sleep studies. This segmented signal is used as the sleep stages signal. Additionally, noise rejection and normalization using the Z-score method are also performed to eliminate noise and eye movements in the signal.

After preprocessing the raw EEG signal, it is then subjected to feature extraction using the proposed two-scale feature extraction module. This module is designed to capture both high- and low-level features in the signal. This module is composed of two branches: the fine feature branch and the coarse feature branch. The fine feature branch is designed for high-level feature extraction using adaptive Kolmogorov convolutional layers. This is useful for capturing high-level features in the signal, such as sleep spindles and rapid eye movements. The coarse feature branch is designed for low-level feature extraction using large kernel sizes for capturing low-frequency features in the signal. This is useful for capturing low-level features in the signal, such as delta waves. The EEG signal is processed through both branches simultaneously to capture a rich feature set containing both high- and low-level features.

The two feature streams are then merged through a cross-scale fusion mechanism. This brings the fine and coarse features together. A regularization mechanism is then used to ensure the model generalizes well and does not overfit. The feature streams then enter the temporal modeling block, where flash attention heads and a lightweight transformer encoder work in tandem. This mechanism digs deep into the time structure of the EEG sequence and brings out the dependencies between different time segments. Flash attention is an efficient mechanism that zeroes in on the most relevant part of the EEG signal. The features are then fused and passed through the lightweight transformer encoder, which refines the features and learns the sequential patterns that occur during the transition between different sleep stages. In order to enhance the clarity and validity of the findings, the approach proposes the integration of attribution-based feedback via the integrated gradient (IG) approach. The integrated gradient approach brings out the most significant part of the electroencephalogram (EEG) signal while pointing at the decision-making process adopted by the model. The temporal features are then subjected to a softmax classification procedure to generate the probability distribution of the various sleep stages. The ultimate results obtained from the classification of the various sleep stages are derived from the transformer attention encoder. The five stages of sleep that have been successfully classified by the model are Wake, N1, N2, N3, and REM.





VI. MODULES DESCRIPTION

EEG Dataset Module

The EEG dataset module provides the raw data for the study. The data are publicly available and include data sets such as Sleep-EDF and SHHS. These data sets contain overnight recordings of brain activity during sleep. The data include the EEG signal and other relevant metadata. These data are the input for the proposed system and provide the required information for training the proposed system for the classification of the sleep stage.

Preprocessing Module

The preprocessing stage processes the raw data obtained from the EEG dataset module. This stage prepares the data for analysis by improving the quality and cleaning the data. The data are passed through a band-pass filter. This filter removes the unwanted noise and other interfering signals. It allows the important frequency band of the signal to pass through. After this stage, the data are segmented into epochs of 30 seconds each Each epoch represents a single data sample corresponding to a particular stage of sleep. After this stage, the data are passed through an artifact rejection filter. This filter removes unwanted data corresponding to eye movements and other unwanted noise. Finally, the data are normalized by the Z-score normalization technique. This technique normalizes the data by bringing the data distribution to a standard normal distribution.

Dual-Scale Feature Extraction Module

The dual-scale feature extraction module is designed for the extraction of the entire spectrum of spectral information contained in the EEG signal. This is done by examining the signal at two scales. The dual-scale feature extraction module consists of two feature extraction branches: the fine feature extraction branch and the coarse feature extraction branch. The fine feature extraction branch consists of convolutional layers with small kernel size and adaptive Kolmogorov convolutional layers. This branch is used for the extraction of the high-frequency characteristics of the signal. This branch is important for the extraction of the subtle characteristics of the signal. These characteristics include the characteristics of the sleep spindle and other short-term characteristics. The coarse feature extraction branch consists of convolutional layers with large kernel size. This branch is important for the extraction of the low-frequency characteristics of the signal.

Cross-Scale Fusion and Regularization Module:

After extracting both fine-scale and coarse-scale features, the system fuses both of these features. This is done by a cross-scale fusion. This module fuses both of the features extracted by the previous module. This fusion results in a single, unified feature representation of both fine-scale features and coarse-scale features. During the entire process, the system also employs a series of regularization techniques to prevent overfitting. This is done to allow the model to perform better on unseen data. The fused features are then used for the subsequent temporal analysis and final classification.

Temporal Attention Modeling Module:

The stages of sleep occur one after another. The current sleep stage is a result of the previous sleep stage and the upcoming sleep stage. To account for this, the system employs a temporal attention modeling module. This module employs flash attention heads and a lightweight transformer encoder. This module is responsible for understanding the temporal dependencies of the EEG signal. This module employs the flash attention mechanism, which is an efficient attention mechanism. This attention mechanism allows the model to assign importance to different segments of the signal. This is done to allow the model to concentrate on the relevant segment of the signal.

Attribution-Guided Feedback Module:

The system also has an attribution-guided feedback module based on the concept of Integrated Gradients. This module highlights the significant EEG signal components that influence the model's prediction. This helps the researcher understand the interpretation of the EEG pattern by the model for labeling the sleep stage. This increases the reliability of the system.

Classification Module:

The final part of the system is the classification module. This module outputs the predicted sleep stage based on the processed features. The enhanced feature representation obtained in the previous temporal modeling module passes through a fully connected neural network layer and a softmax activation function. The output of the classification module is the predicted probability of the five possible sleep stages: Wake, N1, N2, N3, and REM. The predicted stage with the highest probability is the output. This output can be used to generate a full hypnogram for the given EEG signal.

VII. METHODOLOGY

- We are in the process of developing an efficient, automated framework that classifies the stages of sleep by a separated process of extraction from the raw data of the brain through single channel EEG signals. This process is divided into several stages: data gathering, data cleaning and preparation, feature extraction, feature combination, modelling over time, and classification, with each step fine-tuned to refine the representation of the EEG signal as well as the accuracy of the classification process.

- **Data Gathering**: The EEG signals are obtained from publicly available sleep data sets that contain polysomnography recordings performed during the night with sleep stages annotated. The signals obtained are brain activity during sleep, which is necessary for identifying the stages of sleep. The data sets are available with a specific sampling rate, sleep stages, and the corresponding labels, which are the main input for the process.

- **Data Cleaning and Preparation:** The raw data obtained from the data sets are pre-processed for the analysis of the signals. The raw data obtained from the data sets contain unwanted components like muscle, eye, and environmental artifacts, which are eliminated using a band-pass filter, typically a Butterworth filter, which removes the unwanted components while preserving the important components of the EEG signals. The pre-processed data is divided into epochs of 30 seconds, as per the American Academy of Sleep Medicine (AASM) standards, with each epoch according to the particular sleep stage, followed by rejection of an artifacts, which removes mainly corrupted data, and the normalization using the Z-score method, which stabilizes the learning process.

- **Feature Extraction:** With the data pre-processed, the system now tries to find the rich patterns in the data from the different frequency bands, which are associated with the stages of sleep. For this, the dual-scale feature extraction strategy is used, which is based on the dual-scale structure consisting of the fine feature branch and the coarse feature branch, with the fine feature branch using small kernel size CNN layers for the extraction of fine features, which are used for the detection of small-scale events like sleep spindles, while the coarse feature branch uses large kernel size CNN layers for the extraction of coarse features, which are used for the detection of large-scale events like delta waves, which are associated with deep sleep.

- After feature extraction, the system performs cross-scale feature fusion, where the outputs from the fine and coarse branches are combined to create a unified feature representation. This fusion process integrates detailed and contextual EEG information, allowing the model to better distinguish between different sleep stages. During this stage, regularization techniques are applied to reduce overfitting and improve the sleep models which were having the ability to make them generalize the stages across various stages and recordings of the sleep patterns and sleep stages.

- The combined feature representation is then processed by a temporal modelling module, which disentangles the signal. Sleep stages occur over a period of time, and hence ,it was such an important task for the model to get understand how the flow of one stage of sleep is into the another sleep stages. This is achieved by employing attention-based temporal modeming, where flash attention heads are combined with a lean transformer encoder.

The flash attention helps in identifying the most important portions of the signal. These portions are then given greater weightage, as reflected by the greater weight assigned to the features. The lean transformer encoder is then employed to understand the evolution of the features over consecutive EEG epochs. This helps in accurately identifying sleep stages, as transitions from one sleep stage to another can be easily identified.

To make the approach more interpretable, attribution-guided analysis is also conducted. This is achieved by employing Integrated Gradients. This helps in understanding which portions of the EEG signal are primarily responsible for the classification. This is an important aspect, as it helps in understanding the decisions being taken by the model. This is particularly useful in the healthcare domain, where it is important for the decisions being taken by the model to be easily understandable.

Finally, the improved features are processed by a soft max classifier, which then predicts the sleep stages for the given EEG signal. The probabilities of sleep stages which are wake, N1, N2, N3, N4, N5 are probably calculated by using the required models and the stages of sleep are selected by the one stage which has high probability and a good score. This is then combined to form a sleep hypnogram, which represents the sleep architecture of the individual.

VIII. RESULT & OUTPUT

The benefit from the system is the ability to accurately identify the sleep stages based on the classification using the EEG signal. Classification of the sleep stages can be achieved by setting up the system after which training using the available EEG signal data set is done. Sleep data sets including labeled EEG signals have been used to evaluate the system's capability to classify sleep stages.

The preprocessing step helped in cleaning up the EEG signal data by filtering out noise and artifacts using a band-pass filtering method and normalization. Once the signal is cleaned up, it is segmented into 30-second segments or epochs for the deep learning system to analyze the signal for specific features tied to sleep stages. The two-scale feature extraction method helped in identifying the details in the signal as well as the overall patterns in the signal. This helped the system identify specific features in the signal as well as overall patterns in the signal.

These features were subsequently fed into the temporal modeling module, which employed the attention mechanism along with the Transformer Encoder to evaluate the features in the signal. This facilitated learning by the system on how the transitions happened among different sleep stages. This helped the system in classifying sleep stages based on the features present in the signal. The system was able to classify the sleep stages based on these features even when there was confusion about similar features in sleep stages like N1 and N2.

The final step in the classification of sleep stages is through probabilities that are calculated for each of the five sleep stages obtained during the previous step – Wake, N1, N2, N3, and REM. The final classification for each epoch is chosen through the use of a softmax function, and these decisions are compared with actual labels for accuracy assessment. The effectiveness of the model in its performance can be evaluated by metrics such as accuracy, precision, recall, F1-score, and Cohen's kappa.

The experimental results reveal that high classification accuracy and a high degree of association with actual labels are achieved by the model. The results reveal that incorporating dual-scale feature extraction and attention-based temporal modeling into a system improves its capacity to differentiate between various sleep stages considerably. In addition, using integrated gradients provides insights into which EEG features affect the model's decision-making process most, thereby improving its interpretability.

The system's output is a list of predicted sleep stages corresponding to each EEG epoch. This output may be visualized in a graph called a sleep hypnogram, which represents a graph of sleep stages throughout a night's sleep.

The experimental results reveal that the proposed system is an effective and reliable solution for classifying sleep stages from EEG signals.

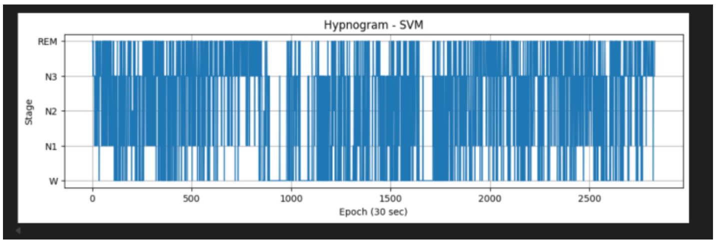


Fig 8.1 Epoch with 30 sec Cycle Using SVM Model

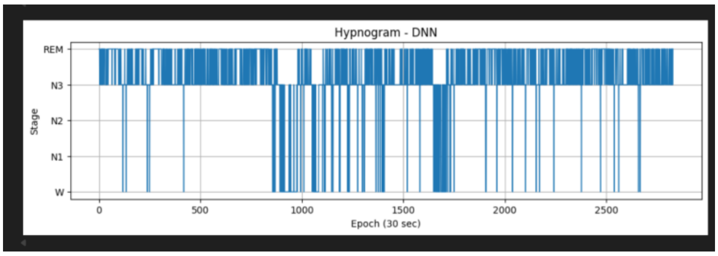


Fig 8.2 Same Hypnogram With 30 Sec Cycle but with a different model

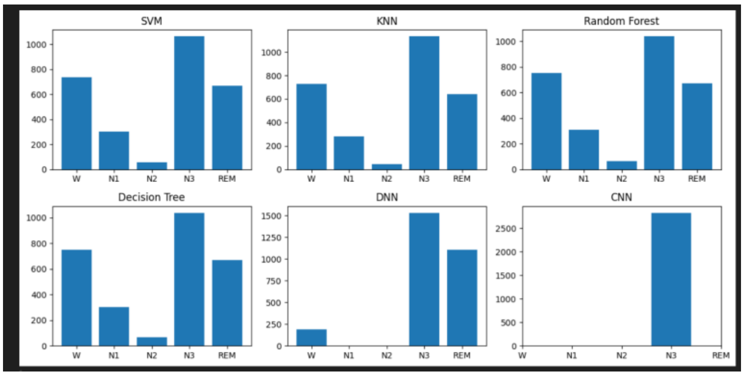


Fig 8.3 Accuracy score of each model which shows which model gains which sleep cycle with highest accuracy

CONCLUSION

The development of an automatic classification system for sleep stages using the EEG signal is done with the aim of increasing the efficiency and accuracy of sleep monitoring systems used to identify various stages of sleep. Sleep stage classification is important when assessing the quality of sleep of an individual, and diagnosing sleep disorders such as insomnia and sleep apnea. Currently, the most widely used methods for classifying sleep stages require the manual review of multiple days' worth of polysomnography data by a trained technician, this can take an enormous amount of time, require large amounts of labour, and be susceptible to error due to human judgment. This study has developed a deep learning framework that is capable of automatically classifying sleep stages based on EEG recordings.

The proposed sleep classification system follows a pipeline that consists of data preprocessing, dual scale (three-dimensional) feature extraction, dual time scale feature fusion, temporal modeming and final classification. In the data preprocessing stage, the EEG recordings were filtered, divided into standard epochs of 30 seconds in length, and normalized to reduce noise and artifacts. In the dual scale feature extraction stage, both waveforms of the EEG recordings were analysed separately for both fine scale and coarse scale characteristics. The classification system's dual scale extraction capabilities will allow for the analysis of both high frequency and low frequency components of the EEG to identify the characteristics of the various stages of sleep.

In order to further enhance the final classification, the extracted features were passed through an attention-based temporal modelling analysis module.

The proposed Hybrid Spectro-Temporal Kolmogorov Attention Framework extends the capabilities of existing single-channel EEG sleep staging models by addressing spectral instability, temporal modeling inefficiencies, and interpretability limitations. By integrating dual-scale Kolmogorov convolution, adaptive cross-scale fusion, efficient transformer attention, and attribution-guided learning, the framework achieves improved robustness and deployment feasibility. Compared to state-of-the-art methods, the system enhances generalization while maintaining high diagnostic accuracy. The architecture demonstrates strong potential for clinical adoption and real-world wearable implementation.